

```

R3
Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 23.0.0.1
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1
R3#
R3#
R3#

```

```

R4
ia - IS-IS inter area, * - candidate
o - ODR, P - periodic downloaded static
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
24.0.0.0/8 is directly connected, Ethernet0/0
24.0.0.2/32 is directly connected, Ethernet0/0
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
45.0.0.0/8 is directly connected, Ethernet0/0
45.0.0.1/32 is directly connected, Ethernet0/0
R4#

```

```

R4#config ter
Enter configuration commands, one per line. End with
4(config)#ip route 172.36.0.0 255.255.0.0 45.0.0.2
4(config)#ip route 12.0.0.0 255.0.0.0 24.0.0.1
4(config)#
Aug 25 18:21:37.914: %PNP-6-PNP_DISCOVERY_STOPPED: PNP
Wizard)
4(config)#ip route 192.50.20.1 255.255.255.0 24.0.0.1
Inconsistent address and mask
4(config)#ip route 192.50.20.0 255.255.255.0 24.0.0.1
4(config)#

```

```

R5
32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/16 is directly connected, Ethernet0/1
L    32.0.0.2/32 is directly connected, Ethernet0/1
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/2
L    45.0.0.2/32 is directly connected, Ethernet0/2
172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.36.0.0/16 is directly connected, Ethernet0/0
L    172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 32.0.0.1
R5#

```

```

R6
12.0.0.0/8 [1/0] via 26.0.0.1
23.0.0.0/8 [1/0] via 26.0.0.1
24.0.0.0/8 [1/0] via 26.0.0.0
26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.2/32 is directly connected, Ethernet0/1
32.0.0.0/8 [1/0] via 26.0.0.1
45.0.0.0/8 [1/0] via 26.0.0.0
172.36.0.0/16 [1/0] via 26.0.0.1
192.50.20.0/24 [1/0] via 26.0.0.1
223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    223.0.50.0/24 is directly connected, Ethernet0/0
L    223.0.50.1/32 is directly connected, Ethernet0/0
R6#

```

```

R2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/0
L    12.0.0.2/32 is directly connected, Ethernet0/0
23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/2
L    23.0.0.1/32 is directly connected, Ethernet0/2
24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/3
L    24.0.0.1/32 is directly connected, Ethernet0/3
26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.1/32 is directly connected, Ethernet0/1
32.0.0.0/8 [1/0] via 23.0.0.2
172.36.0.0/16 [1/0] via 23.0.0.2
192.50.20.0/24 [1/0] via 12.0.0.1
223.0.50.0/24 [1/0] via 26.0.0.2
R2#

```

```

R2
IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/1
L    12.0.0.1/32 is directly connected, Ethernet0/1
23.0.0.0/8 [1/0] via 12.0.0.2
26.0.0.0/8 [1/0] via 12.0.0.2
32.0.0.0/8 [1/0] via 12.0.0.2
172.36.0.0/16 [1/0] via 12.0.0.2
192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.50.20.0/24 is directly connected, Ethernet0/0
L    192.50.20.1/32 is directly connected, Ethernet0/0
223.0.50.0/24 [1/0] via 12.0.0.2
R2#

```



```
R3
Gateway of last resort is 24.0.0.1 for this interface
S   12.0.0.0/8 [1/0]
   23.0.0.0/8 is variably subnetted, 4 subnets, 8 masks
C   23.0.0.0/8 is connected
L   23.0.0.2/32 is connected
S   32.0.0.0/8 is variably subnetted, 4 subnets, 8 masks
C   32.0.0.0/8 is connected
L   32.0.0.1/32 is connected
S   172.36.0.0/16 [1/0]
S   192.50.20.0/24 [1/0]
S   223.0.50.0/24 [1/0]
R3#
R3#
R3#

R4
ia - IS-IS inter area
o - ODR, P - periodic
a - application route
+ - replicated route

Gateway of last resort is 24.0.0.1 for this interface
S   24.0.0.0/8 is variably subnetted, 4 subnets, 8 masks
S   24.0.0.0/8 is distributed
L   24.0.0.2/32 is distributed
S   45.0.0.0/8 is variably subnetted, 4 subnets, 8 masks
S   45.0.0.0/8 is distributed
L   45.0.0.1/32 is distributed
R4#config ter
Enter configuration commands (one per line) and press Enter.
R4(config)#ip route 172.36.0.0/16 24.0.0.1
R4(config)#ip route 12.0.0.0/8 24.0.0.1
R4(config)#
Aug 25 18:21:37.914: %PM-4-ERR: Inconsistent address and mask
R4(config)#ip route 192.50.20.0/24 24.0.0.1
R4(config)#ip route 192.50.20.0/24 24.0.0.1
R4(config)#
```

QEMU (Win-1)

```
My ping statistics for 223.0.50.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=3ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125

Ping statistics for 223.0.50.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=9ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 172.36.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 2ms, Maximum = 9ms, Average = 4ms

C:\Documents and Settings\user>
```

Old Java ASDM

PUTTY

Tftpd32

Start

C:\WINDOWS\system...

8:06 PM

```
U - per-
H - NHR
o - overr

masks
/0
0/0
masks
/2
0/2
masks
/3
0/3
masks
/1
0/1

ea
2

L2 - IS-IS level-2
per-user static route
- NHRP, 1 - LISP

overrides from Pfr

ks

masks
/0
/0
```

R3

Gateway of last resort is 24.0.0.0/8 is variable distance 12.0.0.0/8 [1/0] 23.0.0.0/8 is variable distance 23.0.0.2/32 is directly adjacent 32.0.0.0/8 is variable distance 32.0.0.1/32 is directly adjacent 172.36.0.0/16 [1/0] 192.50.20.0/24 [1/0] 223.0.50.0/24 [1/0] R3# R3# R3# R4

ia - IS-IS inter area o - ODR, P - periodic a - application route + - replicated route Gateway of last resort is 24.0.0.0/8 is variable distance 24.0.0.0/8 is directly adjacent 24.0.0.2/32 is directly adjacent 45.0.0.0/8 is variable distance 45.0.0.0/8 is directly adjacent 45.0.0.1/32 is directly adjacent 4#config ter Enter configuration commands in this mode 4(config)#ip route 172.36.0.0/16 172.36.0.1 4(config)#ip route 12.0.0.0/24 12.0.0.1 Aug 25 18:21:37.914: %PNP-3-ERR: PNP Wizard) 4(config)#ip route 192.50.20.0/24 192.50.20.1 Inconsistent address and interface 4(config)#ip route 192.50.20.0/24 192.50.20.1 4(config)#

QEMU (Win-1)

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASDM

PUTTY

Tftpd32

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASDM

PUTTY

Tftpd32

QEMU (Win-2)

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASDM

PUTTY

Tftpd32

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-I...

Old Java ASDM

PUTTY

Tftpd32

Start

Type here to search

28°C Mostly clear

00:37 26-08-2026

```
QEMU (Win-1)
My Computer
My Network Places
Recycle Bin
Internet Explorer
Cisco ASDM-I...
Old Java ASDM
PUTTY
Tftpd32

C:\Documents and Settings\user>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 172.36.0.10
Pinging 172.36.0.10 with 32 bytes of data:
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=3ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

```
QEMU (Win-2)
My Computer
My Network Places
Recycle Bin
Internet Explorer
Cisco ASDM-I...
Old Java ASDM
PUTTY
Tftpd32

Request timed out.

Ping statistics for 172.36.0.10:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
    Control-C
^C
C:\Documents and Settings\user>ping 172.36.0.10
Pinging 172.36.0.10 with 32 bytes of data:
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 192.50.20.100
Pinging 192.50.20.100 with 32 bytes of data:
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=3ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

Close

R3

```
Gateway of last resort
S 12.0.0.0/8 [1/0]
C 23.0.0.0/8 is va
L 23.0.0.0/8 is
C 23.0.0.2/32 i
L 32.0.0.0/8 is va
C 32.0.0.0/8 is
L 32.0.0.1/32 i
S 172.36.0.0/16 [1
S 192.50.20.0/24 [
S 223.0.50.0/24 [1
R3#
R3#
R3#
```

R4

```
ia - IS-IS inter ar
o - ODR, P - period
a - application rou
+ - replicated rou

Gateway of last resort is
24.0.0.0/8 is variab
24.0.0.0/8 is dis
24.0.0.2/32 is d
45.0.0.0/8 is variab
45.0.0.0/8 is dis
45.0.0.1/32 is d

R4#config ter
Enter configuration comman
R4(config)#ip route 172.36
R4(config)#ip route 12.0.0
R4(config)#
Aug 25 18:21:37.914: %PNP
Wizard)
R4(config)#ip route 192.50
Inconsistent address and
R4(config)#ip route 192.50
R4(config)#
```

QEMU (Win-3)

My Computer, My Network Places, Recycle Bin, Internet Explorer, Cisco ASDM-I..., Old Java ASDM, PUTTY, Tftpd32

```
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms
C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\user>
```

Start, C:\WINDOWS\system...

8:09 PM